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Title: ECMNet: Lightweight semantic segmentation with efficient CNN-Mamba network

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Abstract: In the past decade, convolutional neural networks (CNNs) and transformers have achieved wide application in semantic segmentation tasks. Although CNNs with transformer models greatly improve performance, the global context modeling remains inadequate. Recently, Mamba achieved great potential in vision tasks, showing its advantages in modeling long-range dependency. In this article, we propose a lightweight efficient CNN-Mamba network for semantic segmentation, dubbed as ECMNet. ECMNet combines CNN with Mamba skillfully in a capsule-based framework to address their complementary weaknesses. Specifically, we design an enhanced dual-attention block for a lightweight bottleneck. In order to improve the representation ability of the feature, we devise a multi-scale attention unit to integrate multi-scale feature aggregation, spatial aggregation, and channel aggregation. Moreover, a Mamba enhanced feature fusion module merges diverse level feature, significantly enhancing segmented accuracy. Extensive experiments on two representative datasets demonstrate that the proposed model excels in accuracy and efficiency balance, achieving 70.6% mIoU on Cityscapes and 73.6% mIoU on CamVid test datasets, with 0.87 M parameters and 8.27G FLOPs on a single RTX 3090 GPU platform. Source code will be available at <https://github.com/feixiangdu/ECMNet>.

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